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Course Name: Programming in JAVA

CO1 Coding Challenge

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1. Library Book Management using Classes & Encapsulation

Program:

```
import java.util.Scanner;
class Book {
    private int id; private String title; private boolean issued;
    Book(int id, String title) { this.id = id; this.title = title; }
    boolean issueBook() { if (issued) return false; issued = true; return true;
}
    boolean returnBook() { if (!issued) return false; issued = false; return
true; }
    void display() { System.out.println(id + " " + title + " - " + (issued ?
"Issued" : "Available")); }
}
public class Library {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Book ID: ");
        int id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Book Title: ");
        String title = sc.nextLine();
        Book b = new Book(id, title);
        b.display();
        System.out.print("Issue this book? (yes/no): ");
        if (sc.nextLine().equalsIgnoreCase("yes"))
            System.out.println(b.issueBook() ? "Issued" : "Already Issued");
```

```
        System.out.print("Return this book? (yes/no): ");
        if (sc.nextLine().equalsIgnoreCase("yes"))
            System.out.println(b.returnBook() ? "Returned" : "Not Issued");
    }
}
```

Output:

Status: Accepted

Memory Used: 14860 byte | Execution Time: 0.248 sec

```
Enter Book ID: Enter Book Title: 101 Java Programming - Available
Issue this book? (yes/no): Issued
Return this book? (yes/no): Returned
```

Input

```
101
Java Programming
yes
yes
```

2. Employee Payroll System using Inheritance

Program:

```
import java.util.Scanner;
class Employee {
    String name; double basic;
    Employee(String name, double basic) { this.name = name; this.basic =
basic; }
    double calculateSalary() { return basic; }
}
class PermanentEmployee extends Employee {
    double bonus, pf;
    PermanentEmployee(String name, double basic, double bonus, double pf) {
```

```java
        super(name, basic); this.bonus = bonus; this.pf = pf;
    }
    double calculateSalary() { return basic + bonus - pf; }
}
class ContractEmployee extends Employee {
    int hours; double rate;
    ContractEmployee(String name, int hours, double rate) { super(name, 0);
this.hours = hours; this.rate = rate; }
    double calculateSalary() { return hours * rate; }
}
public class Payroll {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Permanent Employee Name: ");
        String n1 = sc.nextLine();
        System.out.print("Enter Basic, Bonus, PF: ");
        double basic = sc.nextDouble(), bonus = sc.nextDouble(), pf =
sc.nextDouble();
        Employee e1 = new PermanentEmployee(n1, basic, bonus, pf);
        sc.nextLine();
        System.out.print("Enter Contract Employee Name: ");
        String n2 = sc.nextLine();
        System.out.print("Enter Hours Worked and Rate/hr: ");
        int hours = sc.nextInt(); double rate = sc.nextDouble();
        Employee e2 = new ContractEmployee(n2, hours, rate);
        System.out.println(e1.name + ": " + e1.calculateSalary());
        System.out.println(e2.name + ": " + e2.calculateSalary());
    }
}
```

Output:

Output

Status: Accepted

Memory Used: 14960 byte | Execution Time: 0.262 sec

```
Enter Permanent Employee Name: Enter Basic, Bonus, PF: Enter Contract Employee
Name: Enter Hours Worked and Rate/hr: Ravi: 33000.0
Suresh: 40000.0
```

Input

```
Ravi
30000 5000 2000
Suresh
160 250
```

3. Vehicle Rental System using Runtime Polymorphism

Program:

```
import java.util.Scanner;
abstract class Vehicle { abstract double calculateRent(int days); }
class Car extends Vehicle { double calculateRent(int d) { return d * 1500; } }
class Bike extends Vehicle { double calculateRent(int d) { return d * 500; } }
class Bus extends Vehicle { double calculateRent(int d) { return d * 3000; } }
public class Rental {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter vehicle type (car/bike/bus): ");
        String type = sc.next();
        System.out.print("Enter number of rental days: ");
        int days = sc.nextInt();
        Vehicle v;
        switch (type.toLowerCase()) {
            case "car": v = new Car(); break;
```

```
            case "bike": v = new Bike(); break;
            default: v = new Bus();
        }
        System.out.println(v.getClass().getSimpleName() + " rent: " +
v.calculateRent(days));
    }
}
```

Output:

Status: Accepted

Memory Used: 15080 byte | Execution Time: 0.223 sec

```
Enter vehicle type (car/bike/bus): Enter number of rental days: Car rent: 4500.0
```

Input

```
car
3
```

4. Bank Account System using Data Hiding

Program:

```
import java.util.Scanner;
class BankAccount {
    private double balance;
    BankAccount(double balance) { this.balance = balance; }
    boolean deposit(double amt) { if (amt <= 0) return false; balance += amt;
return true; }
    boolean withdraw(double amt) { if (amt <= 0 || amt > balance) return
false; balance -= amt; return true; }
    double getBalance() { return balance; }
}
public class Bank {
```

```java
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter initial balance: ");
        BankAccount acc = new BankAccount(sc.nextDouble());
        System.out.print("Enter amount to deposit: ");
        double dep = sc.nextDouble();
        System.out.println("Deposit " + dep + ": " + acc.deposit(dep) + " 
Balance: " + acc.getBalance());
        System.out.print("Enter amount to withdraw: ");
        double wd = sc.nextDouble();
        System.out.println("Withdraw " + wd + ": " + acc.withdraw(wd) + " 
Balance: " + acc.getBalance());
    }
}
```

Output:

```
Enter initial balance: Enter amount to deposit: Deposit 2000.0: true Balance: 
7000.0
Enter amount to withdraw: Withdraw 100000.0: false Balance: 7000.0
```

Input

```
5000
2000
100000
```

5. University Admission System using Method Overloading

Program:

```java
import java.util.Scanner;
class Admission {
    double calculateFee(String course, int year) { return 50000 + year * 2000; 
}
    double calculateFee(String course, int year, double lab) { return 70000 + 
year * 3000 + lab; }
    double calculateFee(String course, double discount) { return 50000 - 
(50000 * discount / 100); }
}
public class Admissions {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Admission a = new Admission();
        System.out.print("Enter UG course and year: ");
        String ugCourse = sc.next(); int ugYear = sc.nextInt();
        System.out.println("UG Fee: " + a.calculateFee(ugCourse, ugYear));
        System.out.print("Enter PG course, year, lab fee: ");
        String pgCourse = sc.next(); int pgYear = sc.nextInt(); double lab =
sc.nextDouble();
        System.out.println("PG Fee: " + a.calculateFee(pgCourse, pgYear,
lab));
        System.out.print("Enter Scholarship course and discount %: ");
        String schCourse = sc.next(); double discount = sc.nextDouble();
        System.out.println("Scholarship Fee: " + a.calculateFee(schCourse,
discount));
    }
}
```

Output:

```
Enter UG course and year: UG Fee: 52000.0
Enter PG course, year, lab fee: PG Fee: 78000.0
Enter Scholarship course and discount %: Scholarship Fee: 37500.0
```

Input

```
BTech 1
MTech 1 5000
BTech 25
```

6. Shape Area Calculator using Abstract Class & Polymorphism

Program:

java

```
import java.util.Scanner;
abstract class Shape { abstract double area(); }
```

```
class Circle extends Shape { double r; Circle(double r) { this.r = r; } double
area() { return Math.PI * r * r; } }

class Rectangle extends Shape { double l, w; Rectangle(double l, double w) {
this.l = l; this.w = w; } double area() { return l * w; } }

class Triangle extends Shape { double b, h; Triangle(double b, double h) {
this.b = b; this.h = h; } double area() { return 0.5 * b * h; } }

public class ShapeCalc {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius of Circle: ");

        Shape circle = new Circle(sc.nextDouble());

        System.out.print("Enter length and width of Rectangle: ");

        Shape rect = new Rectangle(sc.nextDouble(), sc.nextDouble());

        System.out.print("Enter base and height of Triangle: ");

        Shape tri = new Triangle(sc.nextDouble(), sc.nextDouble());

        Shape[] shapes = { circle, rect, tri };

        for (Shape s : shapes)

            System.out.printf("%s area: %.2f%n", s.getClass().getSimpleName(),
s.area());

    }

}
```

Output:

```
Status: Accepted
Memory Used: 14240 byte | Execution Time: 0.213 sec

Enter radius of Circle: Enter length and width of Rectangle: Enter base and
height of Triangle: Circle area: 78.54
Rectangle area: 24.00
Triangle area: 20.00
```

Input

```
5
4 6
8 5
```

7. Hospital Management System using Interfaces

Program:

```
import java.util.Scanner;

interface MedicalRecord { void addRecord(String d); void displayRecord(); }

class Patient implements MedicalRecord {

    String name; java.util.List<String> records = new java.util.ArrayList<>();

    Patient(String name) { this.name = name; }

    public void addRecord(String d) { records.add(d); }

    public void displayRecord() { System.out.println(name + ": " + records); }

}

class Doctor implements MedicalRecord {

    String name; java.util.List<String> records = new java.util.ArrayList<>();

    Doctor(String name) { this.name = name; }

    public void addRecord(String d) { records.add(d); }

    public void displayRecord() { System.out.println("Dr. " + name + ": " +
records); }

}

public class Hospital {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Patient Name: ");

        MedicalRecord p = new Patient(sc.nextLine());

        System.out.print("Enter Patient Record: ");

        p.addRecord(sc.nextLine());

        System.out.print("Enter Doctor Name: ");

        MedicalRecord d = new Doctor(sc.nextLine());

        System.out.print("Enter Doctor Record: ");

        d.addRecord(sc.nextLine());

        p.displayRecord();

        d.displayRecord();

    }

}
```

Output:

```
Enter Patient Name: Enter Patient Record: Enter Doctor Name: Enter Doctor Record:
Arun: [BP: 120/80]
Dr. Meera: [Prescribed paracetamol]
```

Input

```
Arun
BP: 120/80
Meera
Prescribed paracetamol
```

8. Online Shopping Order System using Packages

```
import java.util.*;
// In a real project, Product would be in package shopping.product
class Product {
    String name;
    double price;
    Product(String name, double price) {
        this.name = name;
        this.price = price;
    }
}
// In a real project, Order would be in package shopping.order
class Order {
    List<Product> items = new ArrayList<>();
    List<Integer> qty = new ArrayList<>();
    void addItem(Product p, int q) {
        items.add(p);
        qty.add(q);
    }
    double total() {
        double t = 0;
        for (int i = 0; i < items.size(); i++)
            t += items.get(i).price * qty.get(i);
```

```
            return t;
        }
    }
    public class MainApp {
        public static void main(String[] args) {
            Scanner sc = new Scanner(System.in);
            Order order = new Order();
            System.out.print("Enter number of products: ");
            int n = sc.nextInt();
            for (int i = 0; i < n; i++) {
                System.out.print("Enter product name, price, quantity: ");
                String name = sc.next();
                double price = sc.nextDouble();
                int q = sc.nextInt();
                order.addItem(new Product(name, price), q);
            }
            System.out.println("Total: " + order.total());
        }
    }}
```

Output:

```
Enter number of products: Enter product name, price, quantity: Enter product
name, price, quantity: Total: 56000.0
```

Input

```
2
Laptop 55000 1
Mouse 500 2
```

9. Student Result Analyzer using Arrays of Objects

Program:

```
import java.util.Scanner;
class Student {
    String name; int[] marks;
    Student(String name, int... marks) { this.name = name; this.marks = marks;
}
    int total() { int t = 0; for (int m : marks) t += m; return t; }
    double avg() { return total() / 3.0; }
}
public class ResultAnalyzer {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();
        Student[] s = new Student[n];
        for (int i = 0; i < n; i++) {
            System.out.print("Enter name and 3 marks: ");
            String name = sc.next();
            int m1 = sc.nextInt(), m2 = sc.nextInt(), m3 = sc.nextInt();
            s[i] = new Student(name, m1, m2, m3);
        }
        Student topper = s[0];
        for (Student st : s) {
            System.out.printf("%s: total=%d avg=%.2f%n", st.name, st.total(),
st.avg());
            if (st.total() > topper.total()) topper = st;
        }
        System.out.println("Topper: " + topper.name);
    }
}
```

Output:

```
Status: Accepted
Memory Used: 12592 byte | Execution Time: 0.193 sec

Enter number of students: Enter name and 3 marks: Enter name and 3 marks:
Aravind: total=253 avg=84.33
Divya: total=285 avg=95.00
Topper: Divya
```

Input

```
2
Aravind 85 90 78
Divya 95 92 98
```

10. Smart Home Device Controller using Hierarchical Inheritance

Program:

```
import java.util.Scanner;
class Device {
    String name; double power; boolean on;
    Device(String name, double power) { this.name = name; this.power = power;
}
    void turnOn() { on = true; }
    double getPower() { return on ? power : 0; }
}
class Light extends Device { Light(String n) { super(n, 60); } }
class Fan extends Device { Fan(String n) { super(n, 75); } }
class AC extends Device { AC(String n) { super(n, 1500); } }
public class SmartHome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter device type to turn on (light/fan/ac): ");
        String type = sc.next();
        Device d;
        switch (type.toLowerCase()) {
            case "light": d = new Light("Light"); break;
            case "fan": d = new Fan("Fan"); break;
            default: d = new AC("AC");
```

```
            }
            d.turnOn();
            System.out.println(d.name + " power used: " + d.getPower() + "W");
        }
    }
```

Output:

```
Enter device type to turn on (light/fan/ac): AC power used: 1500.0W
```

Input

```
ac
```